

Log File Analyzer Report

Objective

The objective of this project is to analyze application log files and generate a summary report of ERROR and WARNING occurrences grouped by module.

Methodology

1. Reading the Log File

The log file was opened and read line by line using Python file handling functions.

2. Identifying ERROR and WARNING Entries

Each log entry was examined to determine whether it contained the keywords ERROR or WARNING.

3. Extracting Module Names

The module name was extracted using string processing techniques by splitting the log line using the keyword "Module=".

4. Counting Frequencies

A nested dictionary structure was used to maintain counts of ERROR and WARNING messages for each module.

Example structure:

```
{ "Database": { "ERROR": 2, "WARNING": 0 } }
```

5. Generating the Report

The collected data was converted into a Pandas DataFrame and exported as a CSV file named log_report.csv.

Sample Results

Module	ERROR Count	WARNING Count
Auth	0	1
Database	2	0
Storage	1	1

Conclusion

The application successfully parsed log entries, grouped messages by module, calculated error frequencies, and generated a structured CSV report. This approach can be extended to process multiple log files and larger production log datasets.